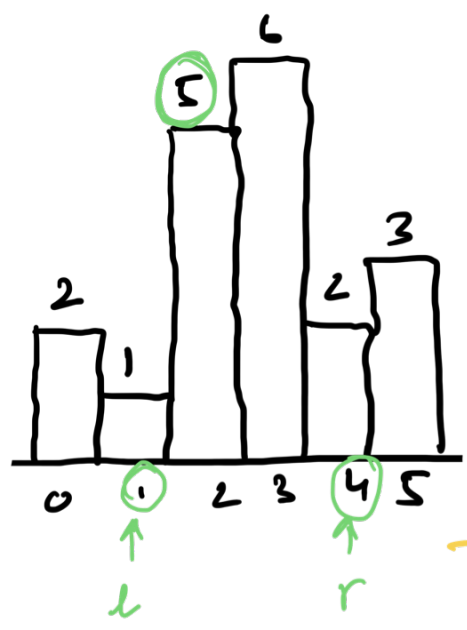
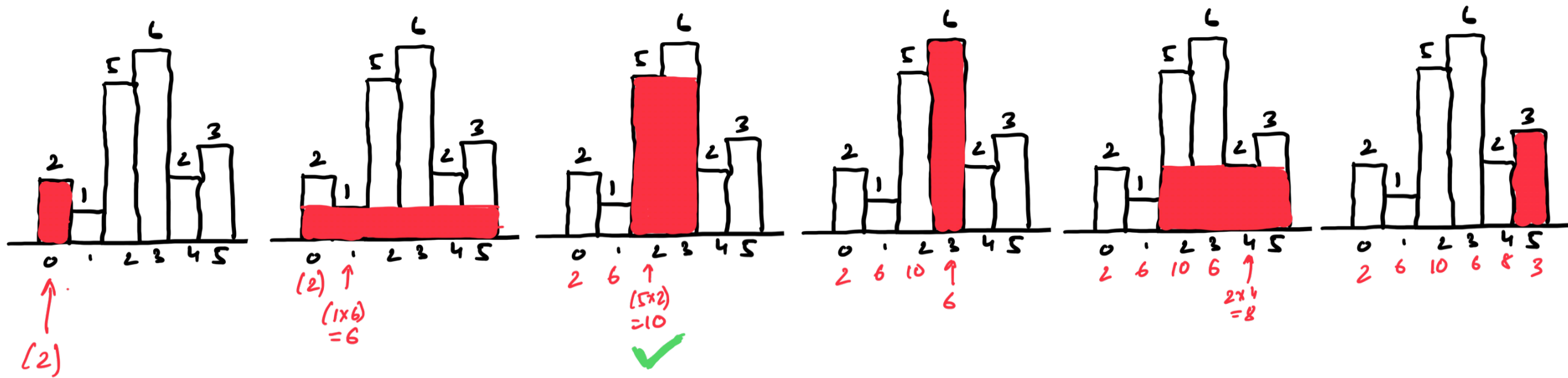
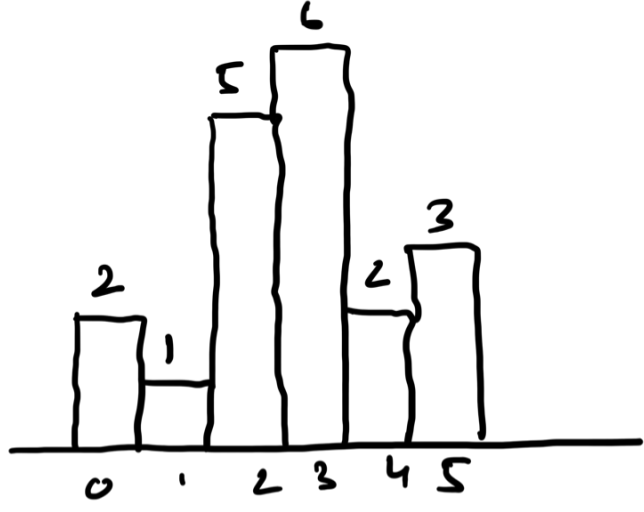


largest rectangle in histogram
heights = [2, 1, 5, 6, 2, 3]



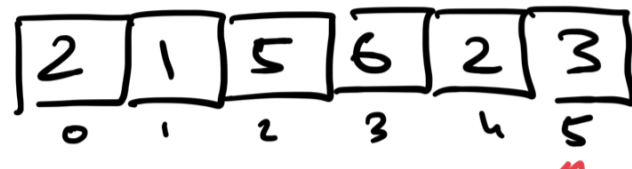
we need to find
left smaller value &
right smaller value to
bound the area

height[i] = 5
width = 4 - 1 - 1 = 2
area = 10

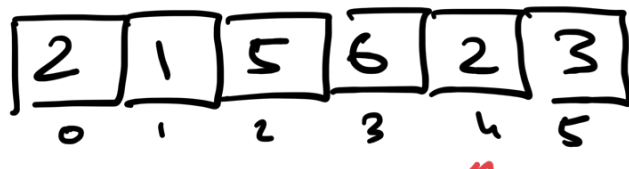
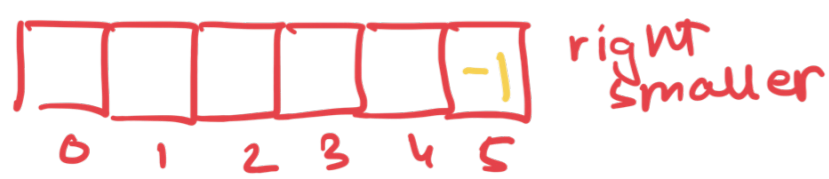
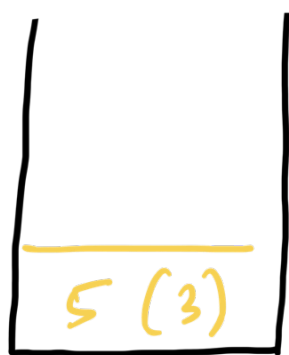
```
ans = 0
for (i = 0 to n) {
    currArea = height[i] * width
    ans = max(ans, currArea)
}
return ans
```

HOW TO FIND RIGHT AND LEFT SMALLER VALUE?

RIGHT

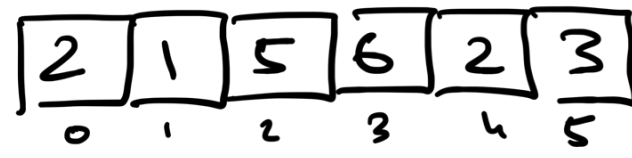
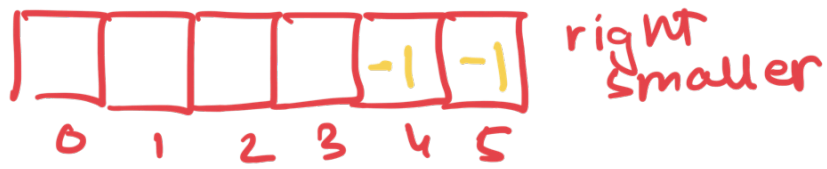
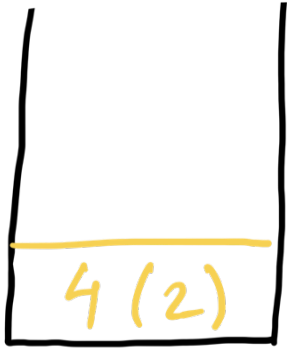


3 ka kuch right boundary
hai hi nahi, write -1
& add in stack

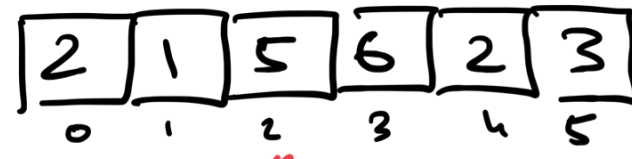
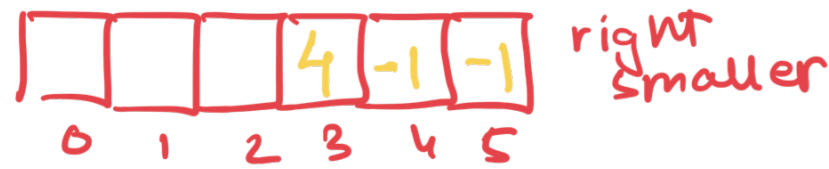
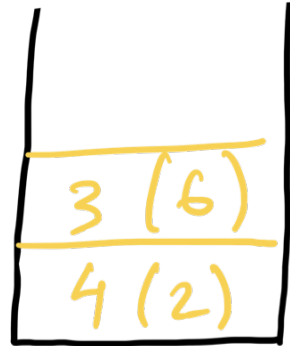


2 itself is smaller
than 3 so can't be
a right smaller
-1 & add in stack

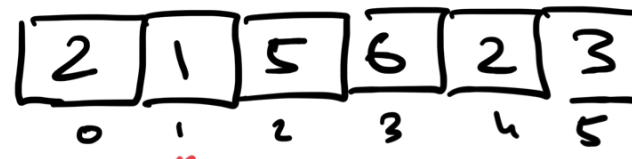
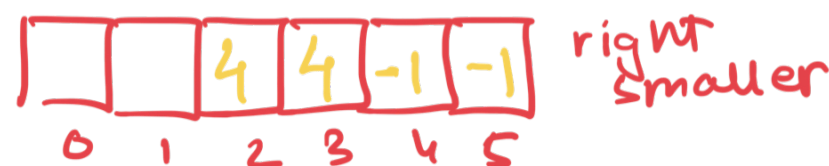
BUT POP
PURANA
FIRST



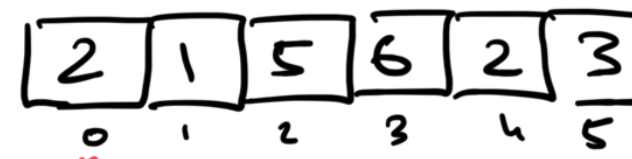
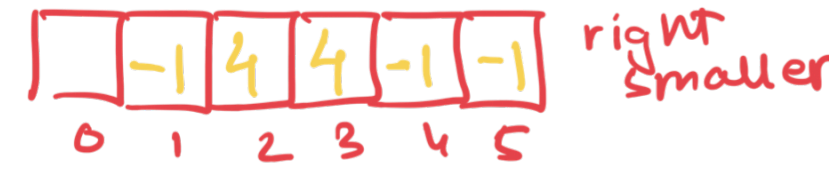
for 6 the immediate right smaller
is 2 which is in stack so write
it's index in the array and stack
index of 6 now don't pop purana
cause its a valid right smaller



6 can't be smaller so pop it
but 2 is smaller, write the index
of that in array and
add index of 5 in stack



both 5 and 2 are greater
there are no valid smaller
pop them both out and add
index of 1 instead



Code

```
left(n, 0)
right(n, 0)
stack <int> s
for (i = n-1; i >= 0; i--) {
    // to find right smaller value
    while (s.size() > 0 && ht[s.top()] >= ht[i]) {
        s.pop()
    }
    right[i] = s.empty() ? n : s.top();
    s.push(i)
}
while (!s.empty()) {
    s.pop();
}
for (i = 0 to n) {
    // to find left smaller value
    while (s.size() > 0 && ht[s.top()] >= ht[i]) {
        s.pop()
    }
    left[i] = s.empty() ? -1 : s.top();
    s.push(i)
}
ans = 0
for (i = 0 to n) {
    width = right[i] - left[i] - 1
    currArea = height[i] * width
    ans = max(ans, currArea)
}
return ans
```